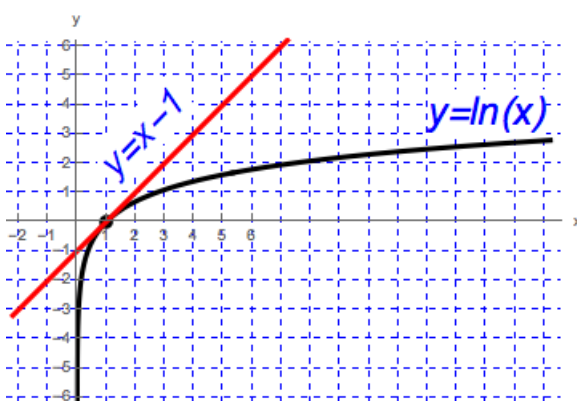


## Problem 7

## Problem 7

**Problem 1** Sketch both the curve  $y = \ln(x)$  and the tangent line to the curve at the point where  $x = 1$ .

Then, write an equation of the tangent line to the curve  $y = \ln(x)$  at the point where  $x = 1$ .



**Explanation.**

In order to write an equation of the tangent line we need to find coordinates of the point  $P(1, \ln(x))$ , and the slope of the line,  $y'(1)$ .

$P(1, 0)$ , and, since  $y'(x) = \frac{1}{x}$ , we get that  $y'(1) = 1$ . Therefore, the equation is given by

$$y = x - 1.$$